A Reconfigurable Digital Compute-In-Memory Heterogeneous Macro for Differential Frame Convolution and Spiking Neural Network

Cheng Zhao[1], Li Lun[1], Zhenhui Dai[1], Haozhe Chen[2], Yingying Cui[1], Xiaoxin Cui[1,2,*]
[1] *School of Integrated Circuits, Peking University, Beijing, China*
[2] *School of Software & Microelectronics, Peking University, Beijing, China*
[*] Email: cuixx@pku.edu.cn

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***Abstract*—The application of artificial neural network (ANN) in video processing encounters significant challenges, including large data volumes, numerous linear operations, and high power consumption. The fusion of convolutional neural network (CNN) and spiking neural network (SNN) provides a dual benefit of achieving high accuracy while maintaining low power consumption. However, ongoing challenges remain in minimizing multiply-accumulate (MAC) operations and optimizing data movement. In this work, we propose a reconfigurable digital compute-in-memory (RDCIM) heterogeneous macro without the sense amplifier, tailored for the diverse computational demands of CNN and SNN. To improve energy efficiency, differential frame convolution (DFC) is adopted to mitigate the computational overhead. In addition, computational resources are functionally reused to accommodate four data flow types, supporting both DFC and SNN operations. Implemented by TSMC 28nm technology, the proposed RDCIM heterogeneous macro achieves the peak energy efficiency of 29.13 TOPS/W for DFC and 0.56 pJ/SOP for SNN, operating at a frequency of 284 MHz.**

Keywords—Heterogeneous Processing, Differential Frame Convolution, Spiking Neural Network, Compute-In-Memory, Dynamic Logic Circuit.

I. Introduction

Video-based target detection remains a pivotal research within the realm of intelligent video surveillance, with the potential to be accelerated through the deployment of convolutional neural network (CNN) on application-specific integrated circuit (ASIC) and field-programmable gate array (FPGA) [1-3]. Although CNN accelerators exhibit high accuracy in video tasks, they face hurdles such as numerous linear operations and high power consumption, presenting considerable obstacles for edge deployment [2]. Inspired from the biological brain, spiking neural network (SNN) has notable advantages of real-time processing and extremely low power consumption [4-5].

In recent years, the fusion of CNN and SNN has leveraged their complementary advantages to pursue high accuracy and low power consumption concurrently [5-8]. For example, both C-DNN [7] and the heterogeneous chip introduced by [8] are hybrid accelerators that combine CNN and SNN functionalities, featuring separate processing units for CNN and SNN tasks. As highlighted in [8], the authors utilize RRAM compute-in-memory (CIM) for CNN and SRAM compute-near-memory (CNM) for SNN, without facilitating resource reuse between CNN and SNN, as illustrated in Challenge 1 of Fig. 1.

As shown in Challenge 2 of Fig. 1, traditional systems like CNN consist of a large number of both linear operations, such as multiply-accumulate (MAC) operations, and nonlinear operations, resulting in substantial computational demands and data movement. Many energy-efficient CNN processors focus on single-image inference tasks [9-11], neglecting the inter-frame similarities in video tasks. Consequently, Yang et al. [12] proposed an inter-frame data reuse CNN accelerator. Compared with traditional CNN, differential frame convolution (DFC) makes full use of the sparsity of differential frame activations to expedite the processing of video tasks. Nevertheless, data movement still needs to be optimized.

Challenge 3 in Fig. 1 describes the high sparsity characteristic of differential frames in video tasks. To quantify the sparsity, we randomly select a subset of action classes from the video database HMDB51 [13]. The results indicate that the sparsity levels of multiple actions all surpass 80%, suggesting that DFC has the potential to achieve an acceleration of more than 5-fold compared to traditional CNN. Therefore, the ongoing issue is how to effectively skip the computations of sparse activations.

In this work, to tackle aforementioned challenges, we propose a reconfigurable digital compute-in-memory (RDCIM) heterogeneous macro without the sense amplifier (SA), custom-designed to meet the diverse computational requirements of CNN and SNN utilizing TSMC 28nm technology. To further reduce MAC operations, DFC is applied to mitigate the computational overhead and optimize data movement. In addition, the RDCIM heterogeneous macro incorporates two distinct types of resource reuse: the reuse of dynamic logic circuit and the reuse of adder groups. The key contributions of this work are summarized as follows:

- We introduce a reconfigurable digital compute-in-memory architecture that supports both sparse MAC operations for DFC and bitwise AND operations for SNN, achieving an effective reuse of dynamic logic circuit and adder groups.
- We adopt the differential frame convolution with a difference and accumulation fusion technique, accelerating MAC operations and reducing computational resources.
- We utilize a zero detector to evaluate the sparsity level of differential frames and skip fully zero-valued activations, thereby improving the overall energy efficiency at the expense of slightly increased area and power of the zero detector.

II. Background And Model

A. Differential Frame Convolution

As demonstrated in Challenge 2 of Fig. 1, the traditional system performs linear function f on image $x(t_1)$ at time t_1, followed by nonlinear function g. However, the computation involved in DFC is composed of three main components: differential operations, sparse MAC operations, and accumulation operations [14]. The entire computational process can be expressed as

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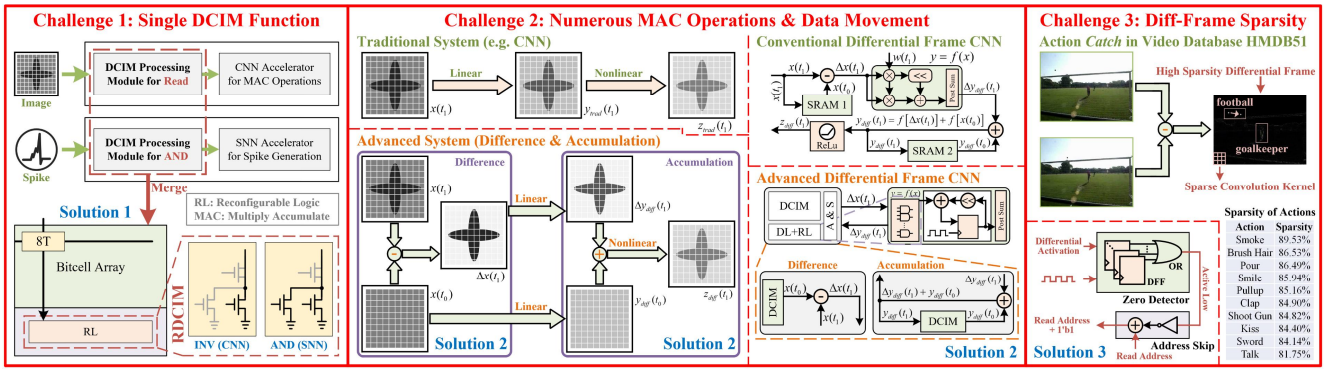


Fig. 1. Motivation and innovation of this work. A&S represents adder & subtracter, and diff-frame represents differential frame.

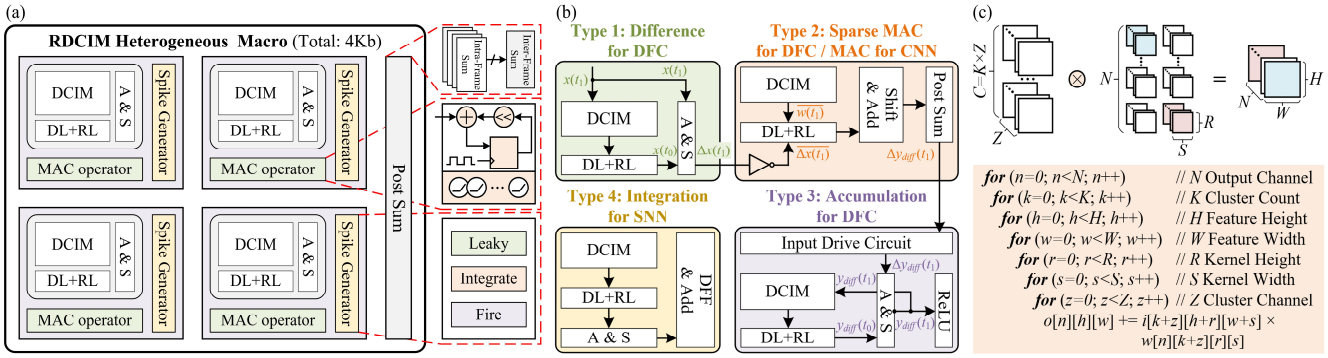


Fig. 2. (a) Overall architecture of RDCIM heterogeneous macro. (b) Four data flow types in RDCIM heterogeneous macro: 1) differential operation for DFC, 2) sparse MAC for DFC/MAC for conventional CNN, 3) accumulation operation for DFC, and 4) integration operation for SNN. (c) Data flow mapping of CNN in RDCIM heterogeneous macro. Input channel C is equal to $K \times Z$.

$$\begin{cases} y_{diff}(t_1) = \Delta y_{diff}(t_1) + y_{diff}(t_0) = f\left[x(t_1) - x(t_0)\right] + f\left[x(t_0)\right] \\ z_{diff}(t_1) = g\left[y_{diff}(t_1)\right] \end{cases}, \quad (1)$$

where $\Delta y_{diff}(t_1)$ is the intermediate result of differential frame $[x(t_1) - x(t_0)]$ executed by the linear function f.

B. Spiking Neural Network

The leaky integrate-and-fire (LIF) model is a widely-used model for simulating the neurodynamics of biological neurons in SNN. The computation of spiking neurons also involves three primary components: integration of synaptic inputs, leakage of the membrane potential, and generation of output spikes [15]. The LIF model can be expressed as

$$\begin{cases} v_l^t[i] = a \cdot v_l^{t-1}[i] \cdot \left(1 - s_l^{t-1}[i]\right) + \sum_j w_l[i,j] \cdot s_{l-1}^t[j] \\ s_l^t[i] = \begin{cases} 1\ , & v_l^t[i] \geq v_{th} \\ 0\ , & v_l^t[i] < v_{th} \end{cases} \end{cases}, \quad (2)$$

where a represents the leaky factor (typically 0.5), w_l denotes the weight in layer l, $v_l^t[i]$ and $s_l^t[i]$ correspond to the potential and spike of neuron i in layer l at time step t, respectively. When the membrane potential exceeds the threshold v_{th}, the neuron fires a spike and resets its potential to zero. Otherwise, the membrane potential gradually diminishes in accordance with the leaky factor.

III. Architecture Design

A. Overall Architecture

The overall architecture of the RDCIM heterogeneous macro is shown in Fig. 2(a), showcasing its capability for heterogeneous processing of DFC and SNN. The RDCIM macro works in either SRAM mode or CIM mode. With a total capacity of 4Kb, the RDCIM macro comprises four RDCIM clusters along with the corresponding intra-frame/inter-frame post-sum module. Each cluster includes three parts: 1) a 1Kb RDCIM array, 2) a MAC operator for DFC, and 3) a spike generator for SNN.

B. Four Data Flow Types

As shown in Fig. 2(b), the RDCIM heterogeneous macro supports four distinct types of data flow. For DFC, it encompasses differential operation, sparse convolution, and accumulation operation (Types 1-3), where sparse convolution for DFC also accommodates the conventional convolution operation. Moreover, Type 4 is dedicated to the integration operation of SNN.

The first type of data flow is the differential operation for DFC. Here, the activations $x(t_0)$ are read from a storage array as four sets of 8-bit data and subtracted from the input activations $x(t_1)$. Subsequently, the input activations $x(t_1)$ are written back into the RDCIM to replace $x(t_0)$.

The second type is the sparse MAC operation for DFC, which decomposes the multiplication of two 8-bit data (i.e., weights and differential activations) into eight sets of 1-bit and 8-bit bitwise AND operations. If differential activations stored in the RDCIM are not sparse, it takes 8 cycles to complete the computation; otherwise, these 8 cycles are skipped to enhance the overall energy efficiency. Following this, the post-sum operation is performed on the data generated by four RDCIM clusters.

Fig. 2(c) presents a visual illustration of the data flow mapping for CNN within the RDCIM heterogeneous macro, supporting various convolution kernel sizes. A single RDCIM cluster is capable of handling the computations for 3×3 convolution kernels with up to 12 input channels without updating weights. In the case that the input channel C of CNN exceeds 12, it is partitioned into two dimensions: K clusters and Z channels in one cluster, where $C=K \times Z$.

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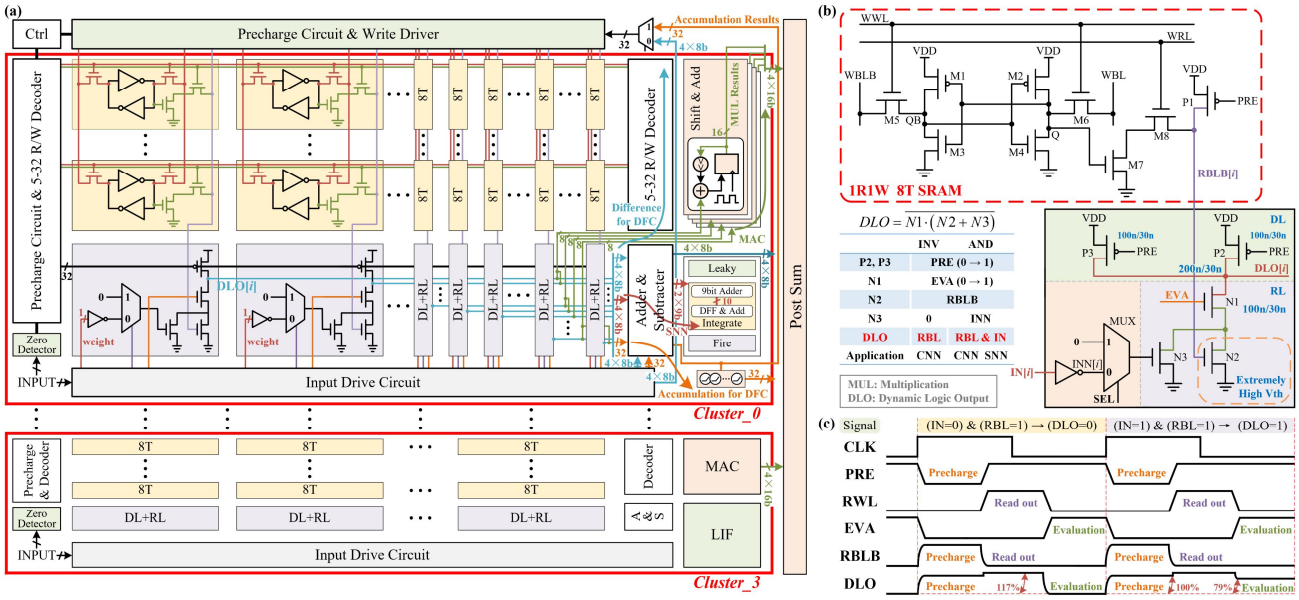


Fig. 3. (a) Detailed description of RDCIM heterogeneous macro. (b) Circuit configuration of an 8T SRAM cell along with dynamic logic circuit, as well as working mechanisms of INV operation and AND operation. In dynamic logic circuit, width-to-length (W/L) ratio of all MOSFETs is 100nm/30nm. Transistor N2 with extremely high threshold voltage is used to minimize read latency. (c) Waveform diagrams for SRAM read operation (i.e., INV operation) and bitwise AND operation under two different scenarios: 1) IN=0, RBL=1; 2) IN=1, RBL=1.

The third type is the accumulation operation for DFC. The MAC results $y_{diff}(t_0)$ are read from a storage array, which are then accumulated with the differential convolution results $\Delta y_{diff}(t_1)$ driven by input drive circuit. Finally, accumulated results $y_{diff}(t_1)$ are written back to the RDCIM, overwriting the previous results $y_{diff}(t_0)$. Simultaneously, accumulated results $y_{diff}(t_1)$ perform the rectified linear unit (ReLU) or other nonlinear functions to serve as the input activation for the subsequent layer.

As illustrated in Type 4 of Fig. 2(b), the RDCIM macro concurrently supports the integration operation for SNN, performing 1-bit and 8-bit bitwise AND operations and accumulating the partial sums. This allows each cluster to perform 4 computations in single instruction multiple data (SIMD) mode. According to (2), spiking neurons also need to carry out leakage (shifting right by 1-bit), reset (setting to zero), and spike generation (comparing with the threshold).

C. RDCIM Macro and Reusable Dynamic Logic Circuit

As shown in Fig. 3(a), a single RDCIM cluster comprises 32×32 8T SRAM array, input drive circuit, precharge circuit, read/write decoder, write driver, adder & subtracter, zero detector, shift & add, and LIF module, all of which satisfy the diverse computational requirements of DFC and SNN.

Fig. 3(b) depicts the circuit configuration of an 8T SRAM cell along with its corresponding dynamic logic circuit. By substituting SA with dynamic logic circuit, the area of readout circuit is reduced by 55.6%, with only a 15.4% increase in read latency as a trade-off. When SEL is 0, DLO is equivalent to the bitwise AND operation between IN and RBL. In contrast, when SEL is 1, DLO represents the voltage state at point Q in the 8T SRAM, implementing the inverter function for the SRAM read operation. By applying distinct signals to the gate of transistor N3, dynamic logic circuit can be reused to realize two different functions.

As depicted in Fig. 3(c), waveform diagrams for the SRAM read operation and the bitwise AND operation are presented. Initially, RBLB is precharged to VDD. Next, if the charge stored at point Q of the 8T SRAM is sufficient to turn on transistor M7, RBLB will be discharged to ground; otherwise, it will gradually decrease to approximately 79% of its initial high voltage state. Lastly, when transistor N2 and N3 reach a stable state, EVA is pulled up to turn on transistor N1, completing the evaluation operation between INN/VSS and RBLB.

D. Reusable Adder Groups

Fig. 4 provides a detailed description of reusable adder groups for both DFC and SNN. Compared with CNN, DFC requires additional differential and accumulation operations, which can be executed by the same four sets of 8-bit adder & subtracter. A single adder & subtracter can support 8-bit signed operations to accommodate negative input activations. When two 8-bit numbers are subtracted, it is essential to monitor for potential overflow, given that differential frame results have an average overflow probability of 0.4% in HMDB51 [13]. For the accumulation operation in DFC, it is necessary to sequentially connect the CI and CO of four sets of 8-bit adder & subtracter to construct a 32-bit adder capable of handling signed multi-bit calculations. In most cases, a 32-bit adder does not encounter overflow issues.

For the integration operation in SNN, two sets of 8-bit adders without overflow judgment are reused as the first level of the adder tree. An additional 9-bit adder acts as the second level of the adder tree to generate a 10-bit accumulated result. This 10-bit accumulated result is then transmitted to the DFF & Add module to complete the accumulation of partial sums.

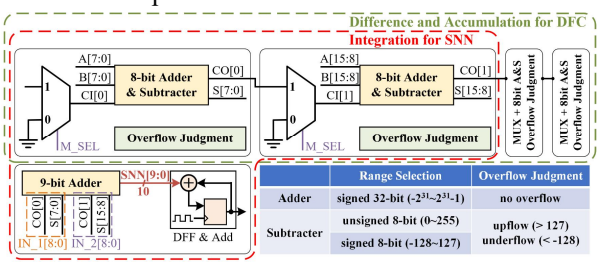


Fig. 4. Illustration of reusable adder groups for both DFC and SNN. Most significant bit (MSB) check is adopted to detect data overflow. Green dashed box represents the group of adders used for difference and accumulation in CNN, while red dashed box represents the group of adders used for integration in SNN. Table shows the range selection and overflow judgment for adders used in DFC.

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TABLE I. COMPARISON WITH PREVIOUS WORKS

	JSSC'24 [7]	ISCAS'24 [16]	JSSC'22 [17]	ISSCC'22 [18]	This work
Technology	28 nm	28 nm	28 nm	28 nm	**28 nm**
Capacity	552KB@SRAM	17Kb@SRAM	64Kb@SRAM	138KB@SRAM	**4Kb@SRAM**
Architecture	Digital Circuit	8T/9T SRAM-Based Digital CIM	6T SRAM-Based Digital CIM	Digital Circuit	**8T SRAM-Based Digital CIM**
Application	CNN & SNN	MAC	CNN	SNN	**DFC & SNN**
Activation Precision	4b/8b/12b/16b@CNN 1-16b@SNN	8b	8b	16b	**8b@CNN 1b@SNN**
Weight Precision	4b/8b	8b	8b	1-16b	**8b**
Frequency (MHz)	50-200	105.6-157.2 (1)	58.1-65.8 (1)	13-115	**284**
Area Efficiency	N/A	0.059 TOPS/mm^2	N/A	N/A	**0.077 TOPS/mm^2@CNN**
Energy Efficiency	24.5 TOPS/W@CNN 1.1 pJ/SOP@SNN	4.61-8.36 TOPS/W	7-7.6 TOPS/W	5.3-12.8 pJ/SOP	**4.21-29.13 TOPS/W@CNN (2) 0.56 pJ/SOP@SNN (3)**

(1) Frequency = 1/Access time; (2) Simulation at 89.53% sparsity (the higher the better); (3) Simulation without sparsity (the lower the better).

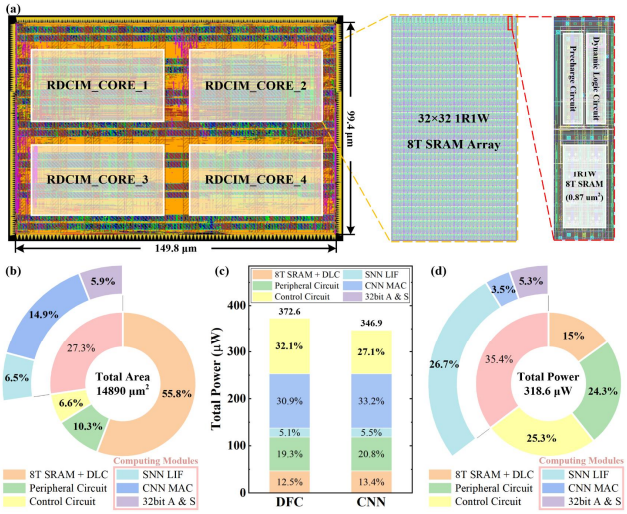


Fig. 5. (a) Layout photographs of RDCIM heterogeneous macro. 1R1W represents 1-read-1-write. (b) Area breakdown of RDCIM heterogeneous macro. DLC represents dynamic logic circuit. (c) Power distribution stack diagram of RDCIM heterogeneous macro for DFC and CNN. The power consumption of 32bit A&S is only 10.76 nW, which can be neglected. (d) Power breakdown of RDCIM heterogeneous macro for SNN.

IV. IMPLEMENTATION AND RESULTS

The proposed RDCIM heterogeneous macro is implemented by TSMC 28nm CMOS High Performance Compact Plus (HPC+) technology. Fig. 5(a) provides a detailed layout overview of the RDCIM macro with a total area of 14890 μm^2, including the layout of an 8T SRAM cell, precharge circuit, and dynamic logic circuit. As shown in Fig. 5(b), the storage array accounts for over half of the entire RDCIM macro's area. Fig. 5(c) and Fig. 5(d) illustrate the power distribution across each submodule within the RDCIM heterogeneous macro under different data flow types. It is notable that the power discrepancy between DFC and CNN is attributed to the zero detector module. For SNN, the LIF module constitutes more than a quarter of the RDCIM macro's power consumption, while the MAC module utilized for CNN is turned off by clock-gating. Therefore, the total power consumption in SNN mode is lower than that of DFC and CNN.

As demonstrated in Table I, the RDCIM heterogeneous macro operates at a clock frequency of up to 284 MHz, reaching the peak performance of 29.13 TOPS/W for DFC under the sparsity conditions of HMDB51 [13]. As illustrated in Fig. 6, once the sparsity level surpasses 12.21%,

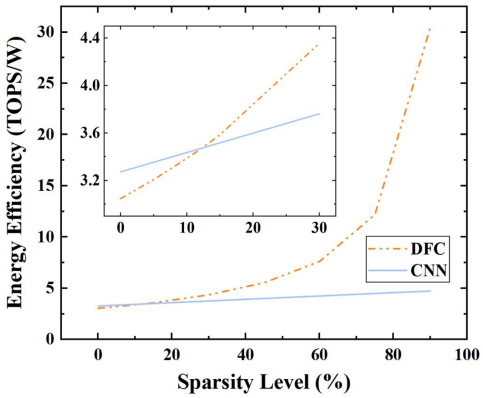


Fig. 6. Energy efficiency curves of DFC and CNN within different sparsity levels.

the energy efficiency of DFC will outperform that of conventional CNN. Moreover, the energy efficiency of DFC increases rapidly with higher sparsity levels. In SNN mode, the RDCIM macro can achieve the energy consumption of 0.56 pJ/SOP with 1-bit activation and 8-bit weight. Consequently, our RDCIM heterogeneous macro can be applied to various neural networks with outstanding energy efficiency without any loss of precision.

A comprehensive comparison with state-of-the-art works is listed in Table I. Benefiting from the effective reuse of computational resources, the RDCIM heterogeneous macro achieves a notable area reduction of over 1417.4 μm^2, accounting for 9.52% of the total area. This optimization enhances the area efficiency to 0.077 TOPS/mm^2 for CNN, marking a 1.31-fold advancement relative to work [16]. These results indicate that our proposed architecture offers improved area efficiency.

V. CONCLUSION

In this work, we propose a SA-less RDCIM heterogeneous macro, custom-designed with TSMC 28nm technology, to meet the diverse computational demands of both DFC and SNN. We utilize a fusion technique of differential and accumulation operations in DFC, with the aim of minimizing computational resources. Furthermore, dynamic logic circuit and adder groups have been functionally reused to operate in four distinct data flow types, supporting DFC, SNN, and conventional CNN. Operating at a frequency of 284 MHz, our RDCIM heterogeneous macro is capable of attaining the energy efficiency of 4.21-29.13 TOPS/W for DFC and 0.56 pJ/SOP for SNN.

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